

# Ganesh Gupta

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## EXPERIENCE

### Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Lead the design and development of backend microservices and RESTful APIs in Java and Python, translating business and technical requirements into scalable application designs for a production system handling millions of daily transactions at 99%+ uptime.
- Built and integrated AI tooling (OpenAI Agents SDK, MCP tool-use, LangGraph, CrewAI, AutoGen) into engineering workflows, applying hands-on experience with LLMs, NLP, and agentic AI to automate business processes across the team.
- Supported cloud and container-based deployment on AWS and GCP (Docker, Kubernetes) with multi-stage CI/CD pipelines, automating build, test, and progressive deployment to reduce manual release overhead.
- Reviewed code, guided technical decisions, and troubleshooted performance and defect issues across the codebase; mentored engineers on code review standards and cloud-native architectural best practices.
- Collaborated with cross-functional teams and project leadership to manage deliverables and track progress against delivery and quality standards.

### Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Led the design and development of 25+ production Java, J2EE, and Spring Boot microservices exposing RESTful APIs for financial transaction processing, serving 1M+ users at 99.9% uptime over 4 years on AWS (EC2, RDS, SQS, Lambda).
- Designed event-driven, Kafka-based asynchronous communication between microservices for real-time financial data processing, ensuring data consistency and fault tolerance across distributed services.
- Standardized REST API contracts, exception handling, and Spring Security/OAuth-aligned authentication patterns across 25+ services, reducing production escalations 60% through consistent retry, timeout, and failure-handling design.
- Applied Core Java (Java 8+) concepts, including Collections, Streams, Lambdas, and strong object-oriented design, to build and refactor high-throughput SQL-backed services; wrote JUnit/Mockito test suites and enforced code coverage standards.
- Led architecture decisions and code-review sessions, mentored engineers on system design, reliability, and secure API design, and established code review standards adopted across the team.
- Partnered directly with enterprise client stakeholders to translate ambiguous business requirements into technical designs, using Agile ceremonies and estimation to manage delivery and track business impact.

## PROJECTS

### LangFetch

[tinyurl.com/LangFetch](https://tinyurl.com/LangFetch)

- Agent decomposing natural-language queries into dependency-ordered SQL subqueries with MCP-compatible tool-use; FastAPI backend deployed as an API server for enterprise data queries.

### Watch-your-LLM

[github.com/ganeshhgupta/watch-your-llm](https://github.com/ganeshhgupta/watch-your-llm)

- Full-stack microservice with FastAPI backend, PostgreSQL, and automated JUnit-style test suites wired into CI/CD, applying object-oriented design patterns for rubric-based evaluation and regression detection.

### RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- Multi-agent system automating production incident diagnosis via tool-use chains, applying AI/NLP concepts and object-oriented design to route alerts to specialized sub-agents and synthesize root cause reports.

## EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

**Languages:** Java (8+), Python, SQL, TypeScript

**Java/Backend:** Spring Boot, Spring MVC, Spring Security, J2EE, RESTful APIs, Microservices, OOP Design, JUnit, Mockito, OAuth

**Messaging & Data:** Kafka, Distributed Messaging, RDBMS/SQL, NoSQL, PostgreSQL, Redis

**Cloud & DevOps:** AWS (EC2, RDS, SQS, Lambda), GCP, Docker, Kubernetes, Git, Jenkins, GitHub Actions, CI/CD

**AI/ML:** LLMs, NLP, Machine Learning, Computer Vision, Agentic AI, OpenAI Agents SDK, MCP, LangGraph

## CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer